PROJECT-BASED LEARNING

PHASE-II EVALUATION

Project Title

Team ID: T294

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T294
Semester: 3rd
Section: ML1,ML2,ML4
Domain: *Smart Water Management*

Team Members

1. *Akanksha Katiyar — 2510378334*
2. *Pragya Anand — 2510011890*
3. *Shivendra Kumar Sahani — 2510370473*

Mentor

Mr.Narayan Chaturvedi

Interactions completed:3 _____

Phase-I Recap & Feedback Incorporated

Phase-I Status

The initial proposal defined the irrigation problem, selected C++ and core data structures, and planned sensor simulation, priority-based allocation, and graph-based routing.

Feedback Received

- *Identified over-irrigation and under-irrigation as key problems.*
- *Planned a graph with Dijkstra's algorithm for water-route calculation.*

Changes Incorporated

- *Implemented 100 farm zones and simulated environmental readings.*
- *Added tank-water checks, full/partial allocation.*
- *Added pipe-network modification and no-route handling.*

Problem Statement & Finalized Objectives

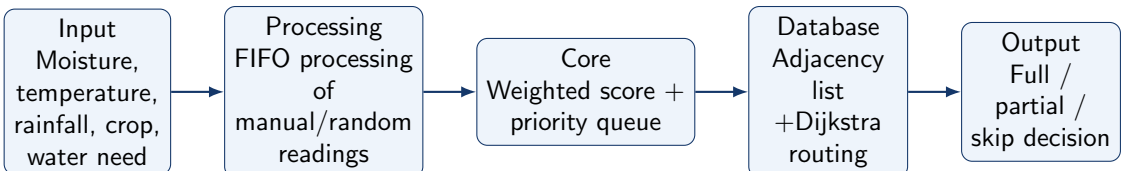
Problem Statement

Traditional irrigation can distribute water unevenly, wasting water in some zones while leaving other zones under-irrigated. The system prioritizes zones based on simulated moisture, temperature and rainfall readings.

Finalized Objectives

1. *Manage data for 100 farm zones, including crop and environmental information.*
2. *Process manual or randomly generated sensor readings through a FIFO queue.*
3. *Calculate irrigation priority and select high-need zones using a priority queue.*
4. *Check water availability and conditions before full, partial, or skipped irrigation.*

Updated System Architecture / Workflow



Major Components

- *Farm Zone Management*
- *Water Tank and Irrigation Management*
- *Graph and Dijkstra Routing*

Changes Since Phase-I

- *Implemented priority-based irrigation.*
- *Added FIFO sensor processing.*
- *Improved irrigation history.*

Module Design & Functional Requirements

Major Modules

1. *Farm sensor module*
2. *Priority module*
3. *Graph module*
4. *Irrigation module*

Functional Requirements

- *Create and manage 100 zones*
- *Accept manual or random readings*
- *Prioritize high-need zones*
- *Add, update and remove pipes*
- *Check water and route availability*

Module Dependency

Sensor readings update zone data → priority logic ranks zones → graph finds route → irrigation module allocates water and records the result.

Technology Stack & Development Methodology

Technology Stack

- Programming: *C++*
- libraries: *iostream, vector, queue, string, iomanip and standard C++ utilities*
- data structures: *vector, queue, priority queue, adjacency-list graph*
- Development Tool: *Dijkstra's shortest path*
- Version Control: *GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Module Development
4. Integration
5. Testing & Validation

Implementation of PBL Course Concepts

Course	Concepts Actually Implemented	Project Module
Data Structures in C	*Queue and FIFO processing for sensor readings; priority queue for selecting zones; vector and adjacency list for network data.*	*Sensor, priority and graph modules*
Algorithms	*Weighted priority calculation and Dijkstra's shortest-path routing.*	*Priority and water-network modules*
C++ Programming	*Structures, standard library containers, modular source files and menu-driven program flow.*	*farm.cpp, graph.cpp, main.cpp*

Development & Implementation Progress

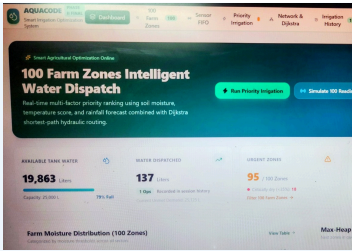
Module	Status	Work Completed
Module 1	Completed	*100 farm zones,Sensor simulation + FIFO queue*
Module 2	Completed	*Priority calculation, Graph + Dijkstra,Pipe management, Water allocation + history*
Module 3	In Progress	*Edge-case testing, Environmental scenarios,Pipe-network testing,Console-output improvements*
Module 4	Pending	*Final screenshots, Documentation checks, Repository/commit details*

Overall Project Completion: *75%*

Working Prototype / Demonstration

Implemented Features

- *Create 100 farm zones, Enter or generate sensor readings*
- *Rank zones by irrigation priority*
- *Find routes with Dijkstra*
- *Manage tank water and view history*



Testing, Results & Validation

Test / Module	Expected Result	Observed Result	Status
Test 1	*100-zone creation*	*100 farm zones are created.*	Pass
Test 2	*Sensor input + FIFO queue*	*Manual/random readings are validated and processed*	Pass
Test 3	*Priority calculation*	*Moisture, temperature and rainfall affect priority.*	*Pass*

Results / Validation

Priority-based water allocation and shortest-route calculation completed,Edge-case testing and final validation ongoing,Core irrigation modules implemented.

GitHub Repository & Development Activity

Repository Information

- GitHub Link: *https://github.com/akankshakatiyar04/Smart-Irrigation-system*
- Contributors: *All Members Added*
- Repository Status: *Updated*

Development Activity

- *Regular commits maintained*
- *Meaningful commit messages*
- *Individual contribution visible*
- *Latest implementation uploaded*
- *Documentation maintained*

Individual Contribution

Member	Role	Technical Contribution	GitHub Commits
Akanksha Katiyar	*Team lead; main.cpp*	*Main menu, program flow, module integration, irrigation execution, water allocation, tank management and history.*	*2*
Pragya Anand	*graph.cpp*	*Adjacency-list graph, pipe addition/update/removal, Dijkstra, route reconstruction and network display.*	*1*
Shivendra Kumar Sahani	*farm.cpp*	*Farm-zone structure, sensor input/simulation, validation, FIFO queue and priority calculation.*	*1*

Challenges, Remaining Work & Phase-III Roadmap

Challenges Faced

- *Designing a weighted priority score*
- *Integrating farm, sensor, graph and irrigation modules*
- *Validating invalid zone/sensor inputs*

Solutions / Learning

- *Test modules individually and together*
- *Expand edge-case and environmental tests*

Phase-III Roadmap

1. *Complete remaining modules*
2. *System integration*
3. *Testing / optimization*
4. *Final deployment / validation*
5. *Documentation*

Expected Outcomes & References

Expected Final Outcomes

- *Complete working prototype / system*
- *Measurable technical results*
- *Practical / industry applicability*
- *Deployment / scalability potential*
- *Research / publication potential*

References

1. *C++ Standard Library documentation*
2. *Standard references on Dijkstra's algorithm*
3. *References for queues, priority queues, vectors and graphs*
4. *Course notes and project documentation*

Thank You
Questions & Suggestions